

Ambient Systems

Machine Learning Technology for Smart Buildings

The Problem

Building owners and operators often struggle to manage the indoor environment in real-time because of three underlying challenges.



Air Quality

Indoor air quality is essential for creating an optimal indoor environment, yet it is difficult to measure in real-time.

Technological Complexity

Building operations consist of complex legacy systems and machinery that controls cooling, ventilation, and other processes.

Competing Priorities

Different buildings have different system requirements based on the unique needs of occupants and operators.

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Air Quality

Indoor air quality needs to be within specified boundaries to create an optimal indoor environment based on unique building operations and occupant needs. It is calculated by measuring three parameters: temperature, relative humidity, and contaminants (including CO_2).

Challenge #1

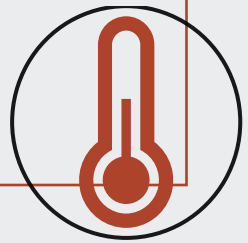
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Air Quality

There are three essential parameters that need to be measured and managed in order to create an optimal indoor environment based on unique building operations and occupant needs.

- Impacts comfort and productivity
- Correlated with humidity
- Energy intensive

Temperature



- Must be within a specific healthy range
- Correlated with temperature
- Impacts spread of contaminants

Relative Humidity



- Organic: pollen, mold, bacteria
- Inorganic: VOCs, SVOCs, Ozone (O_3)
- CO_2
- Other airborne pathogens

Contaminants



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Challenge #2

Technological Complexity

Building operators are increasingly faced with technology integration challenges and difficulties managing operations in-house.



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Competing Priorities

Challenge # 3

Each building has unique operational needs, yet property owners and facility managers are short on in-house skilled talent that can manage and optimize their existing building systems in real-time.



Healthcare

Improve patient outcomes and minimize airborne infections

Offi ces

Increase worker comfort and productivity

Historical Buildings & Museums

Preserve interiors of historical buildings and invaluable artifacts

Indoor Farms

Increase crop yield and streamline operations

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Smart Building Solution

1 Monitoring

Ambient monitors 3 key parameters through a collection of sensors and user feedback

1. Temperature
2. Humidity
3. Contaminants



Sensors + Visibility

2 Recommendations

Ambient provides data-driven recommendations across 3 categories

1. Change temperature
2. Change humidity
3. Increase airflow



Sensors + Visibility Recommendations

3 Automation

Ambient manages 3 groups of devices to automatically improve air quality

1. Thermometers and Humidifiers
2. Purifiers
3. Fans and Ventilators



Sensors + Visibility Recommendations Automation + Actuators

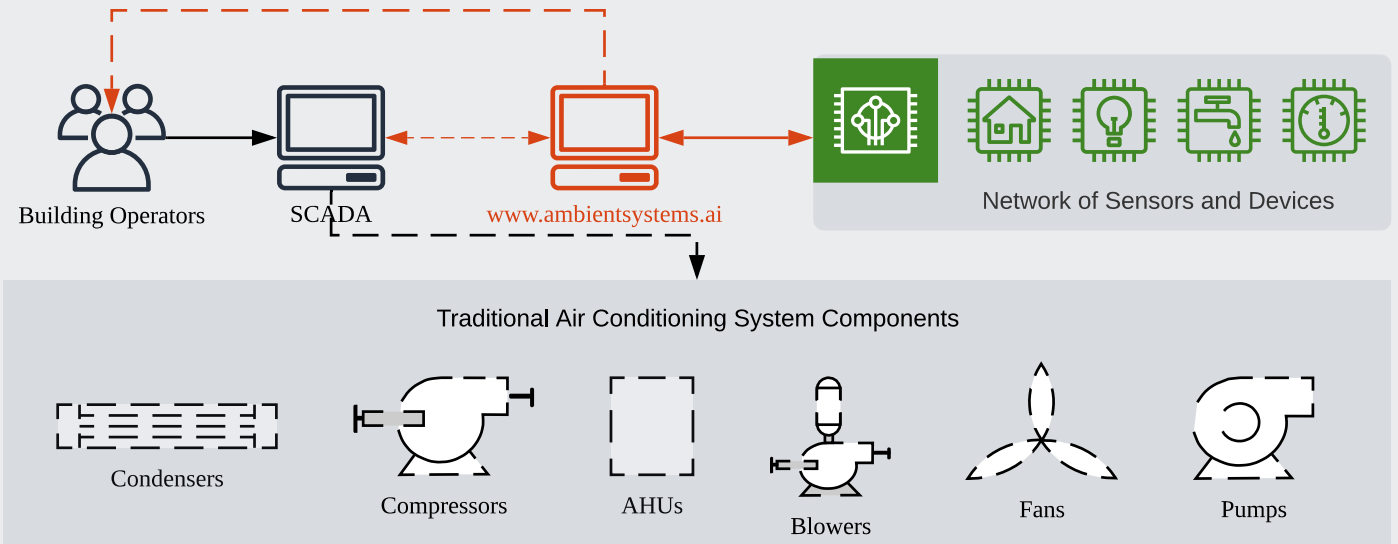
4 Machine Intelligence

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Ambient Systems High-level Schematic

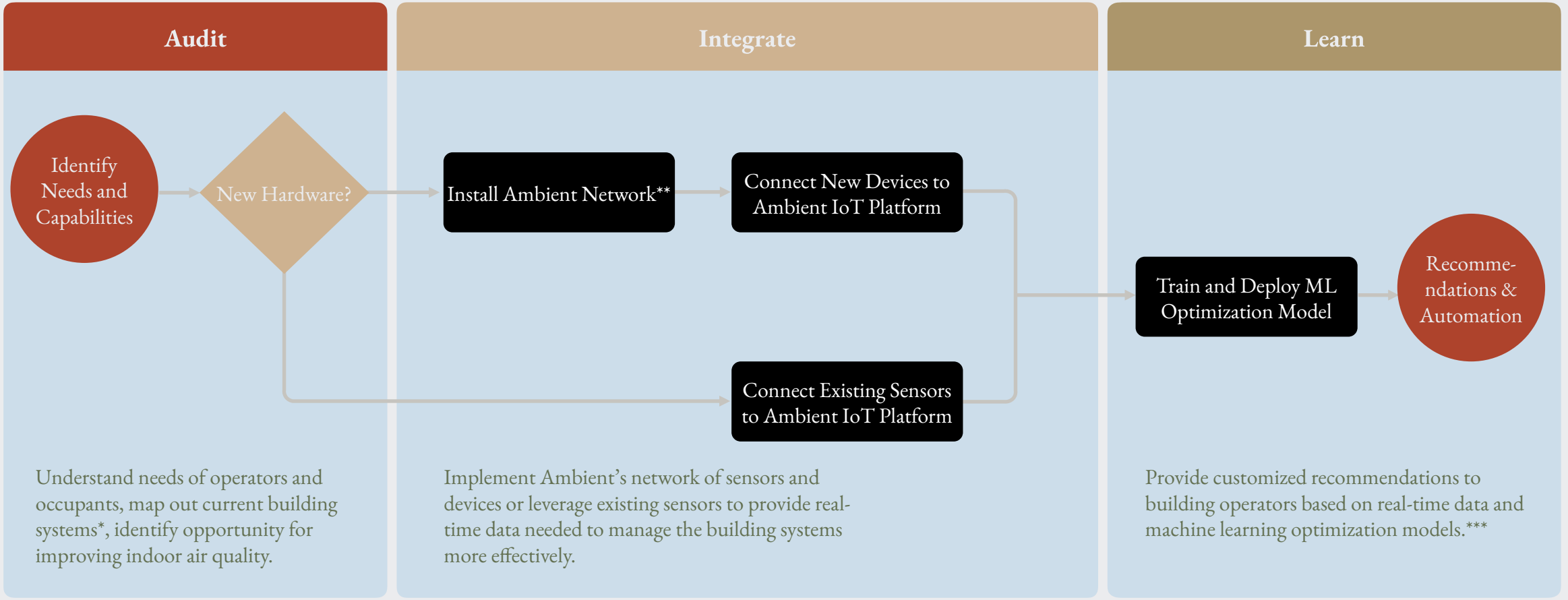
What makes Ambient different?

- Technology tailored for unique building operations and occupant needs
- Seamlessly connects to the building's existing SCADA system and provides notifications directly to the operators
- Adds an additional layer of intelligence using AI-enabled optimization models
- Managed on the cloud and available across your devices
- Enabled by a network of interconnected sensors and devices (Internet-of-Things)
- Leverages NUS intellectual property to minimize maintenance and energy costs of sensors and devices



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Implementation Process



* Focus is on building's systems related to managing indoor air quality, namely mechanical and electrical systems as well as relevant sensors and controls
** Based on unique building needs. Common input parameters: temperature, relative humidity, CO2, contaminants, occupancy, and user feedback.
*** Platform outputs are available online, delivered by email/message notifications, or connected to your building's SCADA system.

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Thank You

Contact

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Appendix

Business Model Canvas

Key Partners

NUS
Property Owners
Facility Managers
HVAC Vendors
Manufacturers
Government Agencies

Key Activities

Building Audits
Platform Development
System Integration

Key Resources

Engineers/Technicians
BD/Sales/Marketing Team

Value Proposition

Integrated turnkey solution for building owners that supports:

- Improved air quality
- Energy efficiency
- Predictive maintenance
- Policy compliance
- Building certifications

Customer Relationships

Dedicated Assistance

Channels

Property Owners
Facility Managers

Customer Segments

<u>Commercial</u>
Hospitals
Offices
Museums

<u>Industrial</u>
Data Centers
Electronics Factories
Cold Storage
Indoor Farms

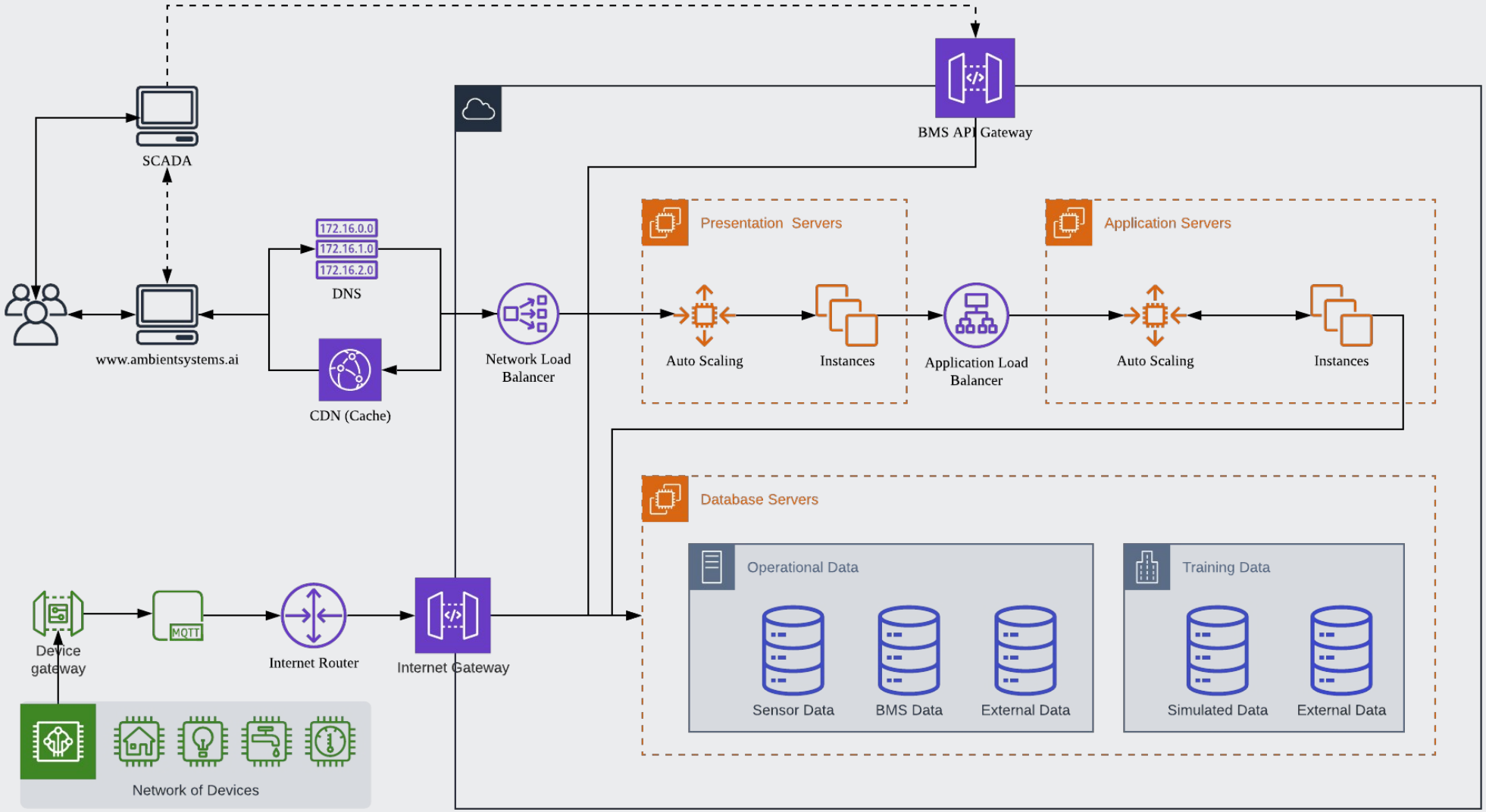
Cost Structure

Revenue Streams

Competitor Analysis – Petal Diagram Smart Cities



Preliminary System Architecture



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Innovation

The core innovation behind the product is the technology integration into a holistic system that leverage cutting-edge intellectual property co-developed with National University of Singapore (NUS).

Component	Tech Layer	IP Opportunity	Advantage
IoT Device	Device hardware	Highly accurate and low-cost sensors	purchase cost • Improve system accuracy
IoT Device	Device hardware	Energy efficient electronic components	operational costs • Increase convenience by charging less often
	Device firmware	New embedded system approaches for processing data at the edge	• Real-time data processing • Lighten load on the cloud
Network Protocols and Processes	Network	A more secure and streamlined approach for connecting IoT devices to the cloud	hardware costs • Improve system performance
Online Application	User Interface	Visibility into multiple sources of data and tailored recommendations for improving the indoor environment	the need for automation • Empower users to improve air quality independently

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Value Proposition Canvas

